

March 14, 2025

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Mr. Faisal D'Souza
National Coordination Office
2415 Eisenhower Avenue
Alexandria, VA 22314

Subject: National Science Foundation Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Dear Mr. D'Souza,

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

The Boeing Company appreciates the opportunity to provide comments to the Office of Science and Technology Policy (OSTP) and the National Science Foundation (NSF) on its "*Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan*". Boeing views the action plan as an important document that can solidify the United States' position as a leader in usable AI-based systems and address key problems for our country.

While the development of AI methods and systems for low-risk tasks and applications has proliferated over the last two decades, developing systems, models and methods that address high-stakes problems is complex, requires significant infrastructure, lacks clear regulatory guidance, and is difficult to validate.

Therefore, our proposal will be centered on technical, mathematical, and infrastructure ideas that will enable our country to reduce the complexity of developing safe and reliable AI-enabled aerospace systems.

1. Real-world Environments for Advancing Research on AI-Enabled Aerospace Systems

We propose the funding and development of real-world and surrogate environments in which AI researchers and engineers can prototype, deploy, test and analyze and test their code, methods, techniques and sub-systems.

We propose the development of state-of-the-art real-world environments for:

- Aviation and aerospace - for advancing research in autonomy and advanced advisory systems for pilots, air traffic controllers, and aviation operators for all phases of flight, building on our experience in developing autonomous prototype

- aircraft that make decisions, advise and perform autonomously during taxi, approach, landing, take-off, and cruise.
- Precision aerospace manufacturing – for advancing research and rapid testing of ideas of manufacturing processes that require high precision, interaction with humans, validation, and certification. In particular, we are interested in integrating perception and reliable human interaction into manufacturing processes for the detection of small objects, advising on anomalous situations, and warning on safety concerns that may occur.

2. Real-World Challenge Problems

Solutions for tasks that require high-stakes decisions and execution in the real-world show a significant acceleration and maturation when challenge problems are set up in the real-world and when the AI system developers are offered a setting in which their systems compete against each other with no supervision from the developers. Real-world problems allow researchers and engineers to test and deploy technologies beyond simulated and lab environments, leading to robust and generalizable technologies that will be transitioned into value-added products. The DARPA Learning Applied to Ground Robots challenge tasks and the DARPA Grand Challenges from the 2004-2008 period opened a path to the development of autonomous cars, through a fast exposure of the research-hard problems that needed to be addressed (such as real-time perception and planning under uncertainty).

In general, for advancing AI methods towards AI engineering methods, prototyping advances much faster when a practically relevant challenge problem (and environment) is addressed, and the research focuses on optimizing constraints that matter for that problem.

We propose the development of challenge problems for decision making in flight, air traffic control, as well as perception and decision making for manufacturing operations.

3. Reasoning for Semantic Understanding

Reasoning – logical or probabilistic – is an intrinsic component of any complex decision-making system and it is central to human intelligence. Reasoning has been at the center of AI research for decades, however, reasoning in high-dimensional spaces (such as the ones that modern machine learning methods excel in) is inefficient.

Reasoning refers to the capability of deriving new information content based on existing information and knowledge. Learned models and generative models alone, can infer concepts only in the regions in which data/observations are available.

Meanwhile, safety critical operations and systems require reasoning and acting in regions with low density data and regions of no data.

Therefore, methods for computationally efficient reasoning are essential for high-stakes applications. Formal probabilistic and logical approaches for reasoning offer semantic interpretations and explainability, which is crucial to safety critical AI engineering applications. Current AI approaches are based on the data-intensive connectionist approach that lacks semantic understanding and rigorous reasoning. Techniques that combine fast (e.g. neural machine learning approaches) and slow reasoning (logical and probabilistic methods), following the approach pioneered by Kahneman [1,2] provide foundational elements for this challenge. Combining probabilistic, logic, and neural approaches, referred to as neuro-symbolic AI, offers the robust way to build semantically sound systems required in safety-critical domains.

Probabilistic reasoning and inference on situations and inputs for which little or no data is available, has received very little attention, and is an important requirement for robust operations of AI-enabled aerospace systems.

We propose additional temporal event reasoning research.

[1] Daniel Kahneman (2011). Thinking, Fast and Slow. Macmillan.

[2] M. Bergamaschi Ganapini, L. Horeh, L. Lamb, A. Loreggia, N. Mattei, F. Rossi, B. Srivastava and K. B. Venable. Proceedings of Thinking Fast and Slow and Other Cognitive Theories in AI. Arlington, VA, AAAI 2022 Fall Symposium.

4. Long-term Memory, Planning and Plan Execution

Insects, birds, and animals with brains of several orders of magnitudes smaller than human brains build nests and store food and return to them months or years after the first time at that location. In an AI-relevant way, these animals can plan and execute plans to reach those locations with tens of millions of intervening and potentially distracting sensory inputs/tokens before recalling these locations and navigating to them. These living analogues compute and operate efficiently in a low-power regime. The construction of such systems would lead to breakthroughs in low-power non-volatile memory computing devices that would be efficiently deployed on edge applications. Further, researching on how to build long-term persistent memory devices that can plan, and act in multi-agent scenarios is highly relevant to future strategic applications in autonomous domains. These devices need to be available and operational in low-connectivity, air-gapped, cloud-independent and constrained environments.

We propose additional research in AI-enabled aerospace systems that have such capabilities. Computationally, these systems will require:

- a. **Memory** – for efficiently summarizing and storing the history of past interaction into a representation and employing it in context;
- b. **Planning** – for devising a set of actions for reaching a long-range goal; and
- c. **Plan Execution** – for choosing the next actions (or tokens) deliberately to achieve the goal or goals safely.

5. Integration of Generative Models into Systems, with Risk Guarantees

The last decade has provided us a wealth of generative AI approaches. These methods can be employed in the design and in the implementation of high-stakes systems, when integrated with other perception and sensing, actuation, expert interfaces.

In particular, generative methods offer a unique opportunity for probabilistically predicting intent and future states of the environment, a function that is required for advanced automation and autonomous systems.

We propose addressing the fundamental research questions that will enable (A) the design of systems that have generative sub-systems and (B) generative models to be integrated into high-stakes systems, such that risk guarantees can be estimated for the performance of the system.

To address these goals, we propose research on extending and developing a formal risk assessment of the system.

6. Robust Human-AI-Enabled System Teams

AI-enabled aerospace systems support complex tasks in conjunction with a human user/expert.

High Reliability Organization (HRO) research (published by Todd LaPorte, Gene Rochlin and Karlene Roberts in the 1996-1999 period) has analyzed and provided the base for robust human team decision making. We can derive the goals for human-computer or human – AI-Enabled System teams that are required to perform in safety critical environments and execute high-stakes tasks.

As human-in-the-loop models become more prevalent in AI technologies, Robust Human-AI systems, developed under the HRO principles can lead to improved technologies for monitoring and oversight needed in complex decision-making scenarios.

We propose additional research on the topics below to advance robust human-computer teaming:

- **Joint Situational Awareness**
- **Detecting and Handling Anomalies**
- **Detecting and Processing Near Misses**
- **Explaining Anomalies and Near Misses**
- **Improvising Solutions**

7. New Mathematics and Novel Fundamental Mathematical and Probabilistic Approaches for Handling and Analysis of Anomalies and Rare Events

Standard statistical methods and approaches currently employed for the analysis of the performance of decision systems fall short in assessing the performance on anomalous observations/situations and extremely rare events. Averages, means, PR and ROC analyses are not sufficient.

We propose the development of a research track that develops novel mathematical and probabilistic metrics and tools for anomalous events.

Thank you for your consideration of our input. Please direct any comments or questions to

[REDACTED]

Sincerely,

Kara Charles
Director, Regulatory Issues Management
Safety, Security & Airworthiness